

REDDING 12 WNW, CA, USA

WMO: 744970

Lat: 40.6507N

Lon: 122.6068W

Elev: 432

StdP: 96.24

Time Zone: -8.00 (NAP)

Period: 03-19

WBAN: 04222

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -2.0	(c) -0.5	(d) -12.6	(e) 1.3	(f) 12.3	(g) -9.9	(h) 1.7	(i) 10.9	(j) 4.9	(k) 7.9	(l) 4.0	(m) 7.1	(n) 1.8	(o) N/A	(p) 0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	39.5	18.5	37.8	18.0	36.0	17.6	20.2	35.3	19.5	34.5	18.9	33.5	1.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.5	22.7	14.2	10.7	23.0	13.2	10.0	23.3	59.6	35.6	57.0	34.2	54.7	33.3	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.0	(b) 3.5	(c) 3.1	(e) -4.2	(f) 42.0	(g) 1.8	(h) 1.6	(i) -5.5	(j) 43.2	(k) -6.5	(l) 44.1	(m) -7.5	(n) 45.0	(o) -8.8	(p) 46.1		
(4) 4.0	(5) 3.5	(6) 3.1	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	7.8	8.6	10.7	13.2	17.8	22.9	26.7	25.7	22.6	16.5	10.7	6.8		
	DBStd	7.90	3.63	3.56	4.15	4.40	4.07	4.41	3.05	2.75	4.01	4.07	4.03	3.15		
	HDD10.0	358	89	63	41	17	1	0	0	0	0	3	38	107		
	HDD18.3	1756	327	272	238	163	59	13	0	1	12	83	230	358		
	CDD10.0	2502	20	24	63	112	243	387	518	486	378	204	59	7		
	CDD18.3	858	0	0	1	9	43	150	260	227	139	27	2	0		
	CDH23.3	11608	2	2	44	177	634	1943	3553	3044	1797	388	23	0		
	CDH26.7	6501	0	0	3	42	236	1048	2200	1827	1007	135	3	0		
(14) Wind	WSAvg	1.8	1.7	1.7	1.7	1.7	1.8	1.8	1.8	1.8	1.9	1.8	1.7	1.7		
(15) Precipitation	PrecAvg	1164	224	177	132	88	61	26	4	7	6	53	142	250		
	PrecMax	1962	671	536	422	218	197	76	19	34	25	147	323	583		
	PrecMin	689	42	19	10	30	2	0	0	2	0	5	32	44		
	PrecStd	356	144	131	107	47	62	25	5	9	8	43	90	167		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	23.2	27.1	30.2	34.6	40.3	41.8	41.0	39.2	33.8	26.4	19.6		
		MCWB	10.2	11.7	13.0	15.1	16.6	19.1	20.0	18.7	17.3	15.8	13.2	9.2		
	2%	DB	19.5	20.5	24.6	27.8	31.6	37.4	40.0	38.6	37.1	30.2	22.6	16.6		
		MCWB	9.1	10.6	12.4	14.2	16.1	17.8	18.9	18.0	16.5	14.5	11.6	8.3		
	5%	DB	16.4	18.0	21.7	25.4	29.4	35.1	38.1	36.7	34.9	27.7	20.0	13.7		
		MCWB	8.3	9.4	11.4	13.4	15.4	17.3	18.5	17.9	16.2	13.8	10.7	7.3		
	10%	DB	13.5	15.1	18.7	22.3	27.1	32.7	36.2	34.9	32.9	25.1	17.0	11.5		
		MCWB	7.6	8.2	10.1	12.2	14.6	16.8	18.0	17.6	16.2	13.2	9.7	7.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	13.0	14.3	17.2	18.6	21.0	21.5	20.6	19.7	17.0	15.1	11.9		
		MCDB	17.1	20.8	22.6	27.1	30.7	35.5	37.9	36.3	33.1	26.2	22.6	14.2		
	2%	WB	10.4	11.9	12.9	15.6	17.4	19.9	20.5	19.8	18.5	15.9	13.7	10.9		
		MCDB	15.4	16.6	20.1	25.8	28.6	34.4	35.9	34.8	32.2	27.1	20.0	12.1		
	5%	WB	9.6	10.9	12.1	14.4	16.4	18.8	19.7	19.1	17.4	15.0	12.1	9.8		
		MCDB	14.0	15.7	18.2	23.7	26.9	33.1	35.2	33.9	31.6	25.7	17.4	11.4		
	10%	WB	8.7	9.9	11.2	13.2	15.5	17.8	19.0	18.3	16.6	14.0	11.1	8.6		
		MCDB	13.0	14.4	16.4	21.5	25.5	31.8	34.1	33.1	30.6	22.8	15.9	11.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.4	11.0	12.3	13.8	15.4	17.5	17.7	16.9	13.8	10.4	8.8		
		MCDBR	14.7	15.5	16.2	17.1	17.8	19.0	19.7	19.7	20.1	18.0	14.3	13.4		
	5% WB	MCWBR	7.4	7.4	7.4	7.5	7.3	6.6	6.7	6.5	7.1	7.0	6.6	7.5		
		MCWBR	11.0	11.8	12.7	14.8	16.3	17.6	18.5	17.7	17.6	15.2	11.6	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.303	0.307	0.319	0.329	0.327	0.334	0.325	0.318	0.313	0.295	0.302		
	taud		2.568	2.562	2.572	2.525	2.531	2.557	2.533	2.549	2.552	2.581	2.602	2.556		
	Ebn at Noon		875	921	952	955	946	945	935	938	926	896	872	838		
	Edh at Noon		70	81	90	101	102	100	102	97	90	79	67	66		
(44) All-Sky Solar Radiation	RadAvg		1.99	2.89	4.10	5.64	7.01	8.00	8.14	7.24	5.76	3.88	2.33	1.66		
	RadStd		0.38	0.41	0.53	0.58	0.51	0.46	0.26	0.26	0.24	0.36	0.27	0.35		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	+0.43	N/A	N/A	N/A	N/A	+101	+17

Nomenclature: See separate page